MIL-G-51349E(EA)
3 April 1987
SUPERSEDING
MIL-G-51349D(EA)
15 November 1974

MILITARY SPECIFICATION

# GRENADE BODY

This specification is approved for use within the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of virgin butyl rubber body for a grenade.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

FEDERAL

| | | |
|---|---|---|
| FED-STD-595 | - | Colors. |
| FED. TEST METHOD STD. NO. 601 | - | Rubber: Sampling and Testing. |

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1330

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-1168 - Ammunition Lot Numbering.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179".)

DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

13-25-72 - Grenade Body.

(Copies of standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)



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ASTM STANDARDS

D 746 - Brittleness Temperature of Plastics and Elastomers by Impact.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. The grenade body shall be in accordance with Drawing 13-25-72 using virgin butyl rubber base compounded to meet the physical properties as specified in table I.

3.1.1.1 Candidate formulation. Prior to submission of first article and without additional cost to the Government, the supplier shall submit the information listed below relating to candidate formulation:

(a) The name of the supplier and the producing subcontractor.

(b) A description of the candidate formulation that will include the following:

(1) The chemical and trade names of the ingredients.
(2) The quantities of each ingredient (parts by weight) including the expected range of variation that can be tolerated without loss of quality in the finished product.

(c) The molding process to be used including:

(1) The mold lubricants, dusting powders, or wet sprays proposed.
(2) Proposed cure times and temperatures.



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TABLE I. Physical properties of molded rubber compound

| Property | As received | Aged, 168 hours at 158° + 1.8°F | Test |
|---|---|---|---|
| 200% modulus | 175 to 300 | 175 to 300 | Test slab 4.4.4.1(a) |
| Tensile strength psi | 1600 min | 1400 min | Test slab 4.4.4.1(b) |
| Elongation % | 600 min | 25% increase 30% decrease | Test slab 4.4.4.1(c) |
| Hardness Shore A | 60 ± 5 | 60 + 10 or -5 | Test slab 4.4.4.1(d) |
| Tear strength lb/in thick | 100 min | 75 min | Test slab 4.4.4.1(e) |
| Flammability | Self extinguish within 10 seconds | | Test slab 4.4.4.1(f) |
| Low temperature | No specimen shall crack or divide upon impact blow following immersion in methanol heat transfer medium at -25°F for 3 + 0.5 minutes | | Test slab 4.3.2.1 |

(d) Assembly process including:

(1) Primers and/or adhesives proposed for use on inserts and center seam.

(2) Proposed cure times and temperatures.

Once the candidate formulation has been approved by the Government, the supplier shall not make any change in the formulation or in the curing and/or assembly processes without obtaining Government approval.

3.1.1.2 Physical properties. Physical properties of the material for the rubber body assembly shall conform to the requirements and tests specified in table I.

3.1.1.3 Color. The color of the material after molding shall be a visual match to either gray, 36231, or green, 34558, FED-STD-595, as specified in the contract or order (see 6.2).

3.1.2 Components. All components shall conform to Drawing 13-25-72 and on the subsidiary drawings.



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3.2 Assembly. The grenade body shall be assembled as shown on Drawing 13-25-72.

3.2.1 Bonding and assembly. The center seam of the body assembly and the bonds of the metal inserts to the upper and lower body hemispheres shall withstand a pull force of 200 pounds minimum when tested in accordance with 4.4.4.3. (Conduct of this test shall be prior to cutting of the emission port hole.)

3.3 Leakage. When tested in accordance with 4.4.4.2, the body assembly without emission port shall not leak, rupture, or separate when subjected to an internal pressure of 20 ± 2 psig.

3.4 Emission port release. When tested in accordance with 4.4.4.4, the tape covering the emission port of the grenade body, when configured as shown on Drawing 13-25-72, shall not release when subjected to an internal pressure of 7.5 ± 0.5 psig.

3.5 Permeability. The weight loss of pyrotechnic mixture within the body assembly shall not exceed 2 percent when tested in accordance with 4.3.2.2.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of bodies and test slabs shall be produced in accordance with this specification for examination and test.

3.7 Workmanship. The body shall be free from imperfections listed in table I of MIL-STD-407 and damage such as rips, tears, or punctures which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.



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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sampling.

4.3.1.1 Test slabs. A quantity of test slabs as specified in table II shall be made for testing. The test slabs shall be made from the same batch of rubber compound as the body halves, using the same methods, materials, equipment, and processes (see Drawing 13-25-72) as will be used during regular production of the body halves.

TABLE II. Preproduction sampling and testing

| Quantity | Requirement | Test |
|---|---|---|
| Test slabs in accordance with FED. TEST METHOD No. 601 | 3.1.1.1 Table I | |
| | 200% modulus | 4.4.4.1(a) |
| | Tensile strength | 4.4.4.1(b) |
| | Elongation | 4.4.4.1(c) |
| | Hardness | 4.4.4.1(d) |
| | Tear strength | 4.4.4.1(e) |
| 75 body assemblies | 3.2 Assembly 3.3 Leak test | 4.4.4.2 |
| 50 body assemblies | 3.2.1 Center seam and insert bonds | 4.4.4.3 |
| 4 test slabs | 3.1.1.1 Flammability | 4.4.4.1(f) |
| 10 body assemblies | 3.5 Permeability | 4.3.2.2 |
| 4 test slabs | 3.1.1.1 Low temperature | 4.3.2.1 |



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4.3.1.2 Body assemblies. A preproduction sample of 15 upper and 15 lower body halves from each cavity shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The body halves shall be assembled into body assemblies in accordance with Drawing 13-25-72, with the exception of the emissions port. The resulting preproduction lot of body assemblies shall be sampled in accordance with table II. If other than Government furnished molds are used, the preproduction lot will consist of 750 body units representing all mold cavities.

4.3.2 Tests.

4.3.2.1 Low temperature. The test slabs shall be tested in accordance with ASTM D 746 (see table I).

4.3.2.2 Permeability. The body assembly shall be filled with CS Riot Mixture (Drawing 143-14-7), or Red Smoke Mixture (Drawing 143-3-9). The threaded openings shall be closed with plugs (Drawing 13-25-76) and adhesive (Drawing 13-25-70) at both ends. For control purposes, one grenade body shall be prepared in the same manner except that the CS Riot Mixture or Red Smoke Mixture shall be omitted. All loaded units and the empty control unit shall be weighed on a suitable balance to an accuracy of 0.1 gram and placed in an air oven maintained at a temperature of 155° ± 5°F. After 30 days, the body assemblies shall be removed from the oven, reweighed and the weight changes recorded. The amount of weight loss of CS Riot Mixture or Red Smoke Mixture shall be determined by subtracting the weight loss of the empty control unit from the weight loss of each bonded unit.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3 to be acceptable. The acceptance number is zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. However, no more than one interfix number of each ingredient in the rubber formulation from one manufacturer shall be contained in any one lot of assembled bodies.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests.

(a) Test slabs. From each batch of raw mix, sufficient test slabs shall be prepared to conduct tests in accordance with table I. Specimens shall be tested as received. The low temperature test is not required.



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(b) Body assemblies. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

(c) Grenade bodies - emission port release. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. The sample grenade bodies shall be examined in accordance with the classification of defects and assigned AQLs.

4.4.3.2 For destructive tests.

(a) Test slabs. The test slabs shall be tested in accordance with table I. The acceptance number is zero.

(b) Body assemblies. The sample body assemblies shall be tested in accordance with 4.4.4.2 and 4.4.4.3 with an AQL of 1.0 percent defective for each test.

(c) Grenade bodies - emission port release. The sample grenade bodies shall be tested in accordance with 4.4.4.4 with an AQL of 1.0 percent defective.

4.4.3.3 Classification of defects.

Grenade body (Drawing 13-25-72).

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | AQL 1.0 percent defective |
| 101 | Component missing or incorrectly assembled |
| 102 | Threads (1-1/4-16UN-2B) incorrect |
| 103 | Rubber cement on interior of body at exhaust port area |
| 104 | Port location or orientation incorrect |
| 105 | Wall thickness incorrect |
| 106 | Tape missing or incorrectly applied |
| 107 | Depth of fuze cavity incorrect |
| 108 | Color not as specified |
| 109 | Workmanship (3.7) |

4.4.4 Tests.

4.4.4.1 Physical properties. The sample test slabs for determining the physical properties of the body assembly material shall be tested as follows:



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(a) Two hundred percent modulus. The slab shall be tested in accordance with method 4131 of FED. TEST METHOD STD. NO. 601 and the tensile stress at 200% elongation determined and recorded.

(b) Tensile strength. The slab shall be tested in accordance with method 4111 of FED. TEST METHOD STD. NO. 601 and the tensile strength at rupture determined and recorded.

(c) Elongation. The slab shall be tested in accordance with method 4121 of FED. TEST METHOD STD. NO. 601 and the elongation at rupture determined and recorded.

(d) Hardness, shore A. The slab shall be tested in accordance with method 3021 of FED. TEST METHOD STD. NO. 601 and the hardness measured and recorded.

(e) Tear strength, die C. The slab shall be tested in accordance with method 4211 of FED. TEST METHOD STD. NO. 601 and the tear resistance determined and recorded.

(f) Flammability. The tip of a test slab (2-1/2 by 1/4 by 1/8 inches) shall be held horizontally at the apex of a 2-1/2 inch blue inner flame of a Bunsen burner operating in a draft free enclosure. After 15 seconds, remove the slab from the flame, hold horizontally and record the time to self-extinguish.

4.4.4.2 Leakage. The body assembly shall be fitted with a suitable adapter and closure plug and tested as follows: Immerse the body assembly in water at room temperature and subject it to the specified pneumatic pressure (see 3.3) for one minute. Units satisfactorily tested shall be randomly selected for testing in accordance with 4.4.4.3.

4.4.4.3 Center seam and insert bonds. Body grenades shall be tested as follows: Suitable threaded adapters shall be installed into the fuze and filling inserts of the body assembly. Using an appropriate tensile test apparatus, the assembly shall be tested at a jaw separation rate of 10 inches per minute. Record the load attained and nature of failure at center seam, or separation of either insert from the rubber sphere.

4.4.4.4 Emission port release. The grenade body, when configured as shown on Drawing 13-25-72, shall be fitted with a suitable adapter and closure plug. The grenade body shall then be subjected to the specified pneumatic pressure (see 3.4) for five seconds. The tape covering the emission port shall not release within this time.

5. PACKAGING

5.1 Unit packing, commercial. The grenade bodies shall be unit packed to provide adequate protection against damage from the supply source to the first receiving activity for immediate use.



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5.2 Packing, commercial. A quantity of items, unit packed as specified in 5.1, shall be packed in containers or special packs as applicable, together with interior packaging, constructed as to give assurance that the items being shipped will remain in prime condition and adequate to assure safe delivery to destination without damage to the contents. The containers shall comply with the Uniform Freight Classification, National Motor Freight Classification, or Department of Transportation regulations, as applicable, and be acceptable for shipment at the most favorable rate. The containers shall be sufficiently rigid to withstand damage to containers or contents. The gross weight and size limitations of the containers shall not exceed the tariff or classification limitations specified therein, and the containers shall be closed and strapped as specified therein.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packages and exterior shipping containers shall be marked in accordance with the domestic marking requirements of MIL-STD-129 to include the following as a minimum:

(a) Noun nomenclature (as cited in the contract or purchase order).
(b) National Stock Number (or part number when NSN is not given or cited in the contract or purchase order).
(c) Government contract or purchase order number.
(d) Quantity.
(e) Contractor's name.
(f) Appropriate address marking (as cited in the contract or purchase order).

6. NOTES

6.1 Intended use. This item is a component of the M47 and M48 grenades. It is intended to contain CS, pyrotechnic mixture, or red smoke mixture.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color of bodies required.
(c) Preproduction:

(1) Time allowed for supplier submission of first article for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.



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6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA - Army Quadripartite Standardization Agreement - 319. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or other appropriate accommodations.

6.5 Subject term (key word) listing.

Hand grenade
Grenade body
Butyl rubber

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A308



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-G-51349E(EA) | GRENADE BODY |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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